

MATSUYAMA, Japan

WMO: 478870

Lat: 33.8435N

Lon: 132.7770E

Elev: 112

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.0	33.7	16.3	12.6	39.5	18.8	14.2	40.5	15.7	41.8	14.1	42.8	4.9	120	0.353

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	92.4	76.9	90.9	76.6	89.3	76.2	78.8	88.4	78.0	87.3	77.2	86.2	7.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.2	137.5	83.5	75.3	133.3	83.0	74.5	129.5	82.7	42.4	88.6	41.5	87.5	40.8	86.5	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.4	11.6	10.0	29.0	95.6	2.0	1.6	27.6	96.7	26.4	97.7	25.3	98.6	23.9	99.7	
			25.5	80.0	2.3	1.1	23.8	80.8	22.5	81.5	21.2	82.1	19.6	82.9	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	43.0	44.2	49.8	58.8	67.3	73.6	81.5	83.3	76.9	67.2	56.7	47.5
	DBStd	14.62	4.04	4.99	5.43	5.48	3.93	4.02	3.64	3.03	4.39	4.94	5.48	4.86
	HDD50	584	220	174	71	4	0	0	0	0	0	7	108	
	HDD65	2782	682	582	472	196	22	0	0	0	32	252	544	
	CDD50	5176	3	12	65	267	536	707	975	1032	808	532	209	30
	CDD65	1897	0	0	0	8	93	257	510	567	358	100	4	0
	CDH74	17100	0	0	1	35	369	1434	5428	6632	2812	386	3	0
	CDH80	6127	0	0	0	2	28	299	2115	2830	812	41	0	0
Wind	WSAvg	4.6	5.0	4.9	5.0	4.8	4.6	4.1	4.5	4.6	4.4	4.3	4.4	4.9
Precipitation	PrecAvg	58.5	2.0	2.8	4.3	4.6	5.9	9.4	8.4	4.7	6.3	4.5	3.1	2.4
	PrecMax	81.9	5.2	5.0	9.3	8.2	11.8	18.3	20.7	13.7	14.9	12.8	6.4	5.6
	PrecMin	31.7	0.2	0.9	1.8	1.9	1.5	1.2	1.1	0.5	2.3	1.3	0.5	0.2
	PrecStd	10.6	1.1	1.1	1.7	1.6	2.6	4.1	5.1	3.5	3.2	3.0	1.5	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	64.9	70.1	78.2	83.0	90.3	93.8	94.9	91.6	84.1	73.9	63.9
		MCWB	50.9	55.7	57.0	63.1	65.6	75.2	77.1	77.0	75.8	72.5	63.5	55.9
	2%	DB	55.1	59.0	65.8	73.7	80.1	86.0	91.8	93.0	88.9	80.6	70.8	60.8
		MCWB	48.2	50.6	54.5	60.1	64.4	73.7	76.7	77.2	75.1	68.9	61.5	52.7
	5%	DB	53.0	55.7	62.5	71.1	77.9	83.3	90.4	91.6	86.9	78.0	68.4	58.3
		MCWB	45.8	48.2	52.6	58.3	63.3	71.2	76.4	77.1	74.1	66.8	59.9	50.8
	10%	DB	50.8	53.0	59.4	68.5	75.8	81.0	88.8	90.2	84.7	75.8	66.1	56.1
		MCWB	44.1	46.5	50.6	56.9	62.6	69.6	76.1	76.7	73.1	65.5	58.5	49.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.8	57.5	60.2	67.4	71.7	76.4	79.9	80.1	78.2	74.8	66.6	59.3
		MCDB	57.0	62.1	65.7	73.2	77.3	84.9	89.5	90.4	86.8	80.7	70.1	61.5
	2%	WB	49.9	52.9	57.0	63.6	68.9	75.1	78.5	79.1	76.9	72.2	64.2	54.9
		MCDB	53.0	57.0	62.4	69.4	74.4	83.4	87.9	89.0	85.0	78.1	68.1	58.4
	5%	WB	47.3	50.2	54.5	61.2	66.8	74.0	77.7	78.3	76.0	69.8	62.0	52.4
		MCDB	51.5	54.4	60.3	67.7	73.0	81.0	86.8	87.9	83.6	75.7	66.9	56.7
	10%	WB	44.8	47.4	52.3	59.3	65.1	72.6	76.9	77.6	74.9	67.4	59.6	50.1
		MCDB	50.0	51.9	58.0	66.0	72.4	78.7	85.8	87.0	82.3	73.3	64.9	55.2
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.3	14.6	15.2	15.1	12.3	12.1	12.9	13.1	14.1	13.9	12.6
		MCDBR	15.2	16.8	18.3	18.5	18.0	14.9	14.1	14.4	14.9	15.9	15.2	15.4
		MCWBR	10.7	11.1	11.5	9.6	8.8	6.3	5.0	5.2	6.2	7.9	9.5	10.4
	5% WB	MCDBR	13.5	14.1	15.4	14.3	13.6	12.0	12.8	13.2	12.4	12.4	12.7	12.9
		MCWBR	11.2	11.0	11.6	9.1	8.5	5.5	5.1	5.2	5.7	7.3	9.4	10.8
Clear-Sky Solar Irradiance	taub	0.408	0.454	0.515	0.545	0.546	0.562	0.561	0.532	0.493	0.449	0.422	0.401	
	taud	2.138	2.036	1.891	1.872	1.934	1.978	2.026	2.111	2.179	2.208	2.194	2.171	
	Ebn at Noon	245	246	241	241	242	238	237	242	246	246	239	239	
	Edh at Noon	39	48	60	63	60	58	55	49	44	40	36	36	
All-Sky Solar Radiation	RadAvg	753	959	1283	1544	1673	1432	1594	1639	1316	1088	812	658	
	RadStd	62	75	96	148	171	164	207	161	131	109	70	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page